

7. Use the set-roster notation to indicate the elements in each of the following sets.

a. $S = \{n \in \mathbf{Z} \mid n = (-1)^k, \text{ for some integer } k\}.$

b. $T = \{m \in \mathbf{Z} \mid m = 1 + (-1)^i, \text{ for some integer } i\}.$

c. $U = \{r \in \mathbf{Z} \mid 2 \leq r \leq -2\}$

d. $V = \{s \in \mathbf{Z} \mid s > 2 \text{ or } s < 3\}$

e. $W = \{t \in \mathbf{Z} \mid 1 < t < -3\}$

f. $X = \{u \in \mathbf{Z} \mid u \leq 4 \text{ or } u \geq 1\}$

a. $S = \{n \in \mathbf{Z} \mid n = (-1)^k, \text{ for some integer } k\}.$

n is an integer of S, such that n equals (-1) to the power of some integer k.

Table of content for possible k integers:

k	-	n
-3	$(-1)^{-3}$	-1
-2	$(-1)^{-2}$	1
-1	$(-1)^{-1}$	-1
0	$(-1)^0$	1
1	$(-1)^1$	-1
2	$(-1)^2$	1
3	$(-1)^3$	-1

S contains two elements: -1 and 1.

$$S = \{-1, 1\}$$

b. $T = \{m \in \mathbb{Z} \mid m = 1 + (-1)^i, \text{ for some integer } i\}$.

m is an integer of T , such that m equals $1 + (-1)$ to the power of some integer i .

Table of content for possible i integers:

i	-	m
-3	$1 + (-1)^{-3}$	0
-2	$1 + (-1)^{-2}$	2
-1	$1 + (-1)^{-1}$	0
0	$1 + (-1)^0$	2
1	$1 + (-1)^1$	0
2	$1 + (-1)^2$	2
3	$1 + (-1)^3$	0

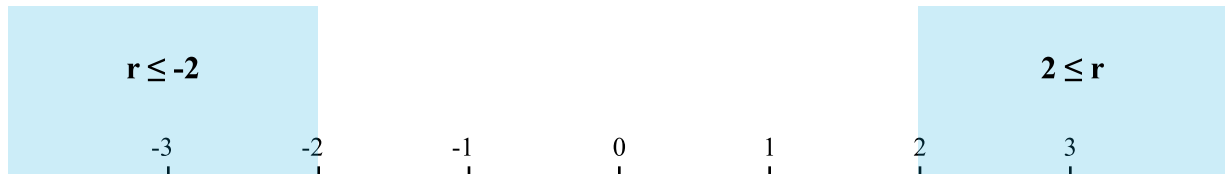
T contains two elements: 0 and 2.

$T = \{0, 2\}$

c. $U = \{r \in \mathbf{Z} \mid 2 \leq r \leq -2\}$

r is an integer of U , such that r is greater than or equal to 2 and less than or equal to -2.

Let's look at the number axis for clues:



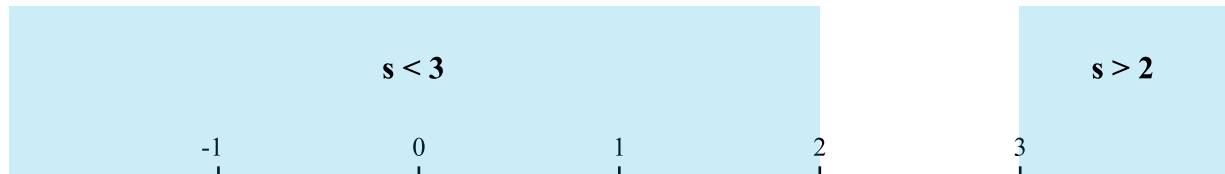
Upon investigating the possible values of r , marked by the blue areas on the number axis, and emphasising the and in the set property, we can conclude that there are no values of r that can fulfil both conditions.

There are no elements.

d. $V = \{s \in \mathbf{Z} \mid s > 2 \text{ or } s < 3\}$

s is an integer of V, such that s is greater than 2 or s is less than 3.

Let's visualise using the number axis again.



Upon investigating the possible values of s, marked by the blue areas on the number axis, and emphasising the or in the set property, while defining s as an integer, we can conclude that all integers are available as possible values of s.

$$V = \{\mathbf{Z}\}$$

e. $W = \{t \in \mathbf{Z} \mid 1 < t < -3\}$

t is an integer of W , such that t is greater than 1 and less than -3.

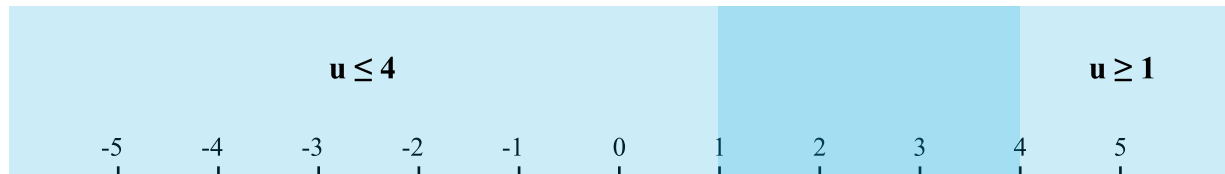
We can use the same reasoning model as we did in c, and further emphasise the and in the set property, to conclude that there are no values of t that can fulfil both conditions.

There are no elements.

f. $X = \{u \in \mathbf{Z} \mid u \leq 4 \text{ or } u \geq 1\}$

u is an integer of X , such that u is less than or equal to 4 or u is greater than or equal to 1.

Let's look at the number axis one more time.



Here the marked areas overlap, and when defining u as an integer, and emphasising the or in the set property, we can conclude that all integers are available as possible values of u .

$X = \{\mathbf{Z}\}$

Solution: **a.** $S = \{-1, 1\}$

b. $T = \{0, 2\}$

c. There are no elements.

d. $V = \{\mathbf{Z}\}$

e. There are no elements.

f. $X = \{\mathbf{Z}\}$